

Gautam Neelakantan Memana

EDUCATION

Australian National University Postdoctoral Researcher	2026-2027 Mentor: Po Lam Yung
University of Wisconsin Madison PhD student PhD thesis: On nonlinear maximally subelliptic PDEs	2021-2026 Advisor: Brian Street
Indian Institute of Science Education and Research, Mohali BS-MS program, Mathematics major MS thesis: L^p Harmonic analysis on Heisenberg group	2016-2021, CGPA: 9.8/10.0 Advisor: Jotsaroop Kaur

PUBLICATIONS/PREPRINTS

- Gautam Neelakantan Memana, “*Comparison theorems for weak solutions of nonlinear maximally sub-elliptic PDEs*”, <https://arxiv.org/abs/2604.12291>
- Gautam Neelakantan Memana, “*A Regularity Theorem For Fully nonlinear Maximally Subelliptic PDE*”, <https://arxiv.org/abs/2409.04344>
- Soma Maity, Gautam Neelakantan Memana, “*Uniform Poincaré inequalities on measured metric spaces*”, Manuscripta Mathematica (2022)
- Gautam Neelakantan Memana, “ *L^p Harmonic analysis on the Heisenberg group*”, MS thesis at IISER Mohali

TEACHING EXPERIENCE

• Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340)	Fall 2025
• Teaching Assistant: Introduction to Discrete Mathematics (MATH240)	Summer 2025
• Teaching Assistant: Calculus and Analytic Geometry II(Calc II)	Fall 2024
• Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340)	Summer 2024
• Instructor: Algebra (MATH 112)	Fall 2023
• Instructor: The Theory of Single Variable Calculus(MATH 421)	Summer 2023
• Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340)	Spring 2023
• Teaching Assistant: Techniques in Ordinary Differential Equations (MATH 319)	Fall 2022
• Direct Reading Program: Ergodic Theory	Fall 2022
• Teaching Assistant: Algebra and Trigonometry(MATH 114)	Spring 2022
• Teaching Assistant: Algebra and Trigonometry(MATH 114)	Fall 2021

SCHOLARSHIPS AND AWARDS

• AMS sectional travel grant	Fall 2025
• Henry Schaerf Mathematics Graduate Award	Fall 2024
• MinneMUDAC best prediction team award in the graduate section	Spring 2025

• DAAD WISE scholarship	May–July, 2019
• VSRP Scholarship for internship at TIFR CAM, Bangalore	May –July, 2018
• Madhava Mathematics competition invitation to attend camp	Summer 2018
• Kishore Vaigyanik Protsahan Yojana scholarship	2017–2021
• INSPIRE Fellowship	2016–2017

INVITED TALKS/PRSENTATIONS

• Workshop OpArt à St Dié des Vosges	May 27, 2026
• AMS special session in Harmonic Analysis- North Dakota State University, Fargo	April 2026
• Contributed talk at AMS Central Sectional Meeting in St. Louis	October 2025
• Cochin University of Science and Tecnology- analysis seminar	May 2025
• Poster presentation at Madison Lectures in Harmonic Analysis	May 2024
• Kopp summer school- Analysis of multiple ergodic averages	September 2023
• University of Florida-analysis seminar	April 2023
• Kopp summer school- Nodal domains and landscape functions	October 2022
• Poster presentation at Nordfjordeid Summer School 2019 - Analysis, Geometry and PDE	June 2019